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RA-11: unrestricted Kronecker trace estimation

Polynomial real simulation, parallel factor access,
and a bounded-probability lower bound

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Prepared with ChatGPT; mathematical arguments and reproducible checks

Status: partial results, not a complete solution of RA-11. The general upper and lower bounds in Theorem 1.1 do not match. A tensor-product subclass is characterized up to constants, and the projection conjecture identified below has an explicit counterexample. The arguments have passed an independent informal Codex AI-agent audit of their stated partial scopes; see the accompanying independent-review.md. This is not external human peer review or formal theorem-prover verification.

Abstract

We study relative trace estimation of arbitrary real positive-semidefinite matrices from adaptive, exact, real rank-one tensor matrix-vector queries. A degree- q interpolation identity simulates one complex product query with $q + 1$ real product queries. It also yields a counterexample, already for local dimension two, to Conjecture 23 in arXiv:2502.08029v2: a fixed span of $q + 1$ real product vectors has squared projection at least 2^{1-q} on every real unit product vector. For the promise $M = A_1 \otimes \cdots \otimes A_q$, with PSD factors of order n , a normalization-and-shift construction provides simultaneous ordinary matrix-vector access to all factors. This gives exact recovery in n original queries and a relative-error estimator using $O(\sqrt{q}/\varepsilon)$ queries. An adaptive Wishart posterior argument, combined with log-concave anti-concentration, gives a matching bounded-probability lower bound for this subclass. Blocking yields a general lower bound, while interpolation and ordinary-query simulation give general upper bounds. The remaining gap for arbitrary PSD matrices is stated explicitly.

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1 Model, source audit, and main results

1.1 The exact target

Fix integers $n, q \geq 2$, $0 < \varepsilon < 1/2$, and $N = n^q$. The unknown input is an arbitrary real symmetric PSD matrix $M \in \mathbb{R}^{N \times N}$. An oracle call specifies $v_1, \dots, v_q \in \mathbb{R}^n$ and returns the *entire* vector

$$M(v_1 \otimes \dots \otimes v_q) \in \mathbb{R}^N.$$

Let $Q(n, q, \varepsilon)$ be the minimum worst-case number of calls needed by a randomized adaptive algorithm whose output T satisfies, for every fixed M ,

$$\mathbb{P}\{|T - \text{tr } M| \leq \varepsilon \text{tr } M\} \geq 2/3. \quad (1)$$

Arbitrary finite exact arithmetic between queries is allowed, and no condition number bound is imposed. These are the quantifiers and oracle specified by RA-11 [1]. In particular, the matrix itself need not be a Kronecker product. Throughout, real-valued Gaussian randomness is used in the usual exact-real oracle model. All displayed query simulations require only finitely many arithmetic operations on the sampled values and responses.

1.2 What the cited sources do and do not supply

Theorem 7 of Meyer–Swartworth–Woodruff [2] concerns a conditioned *scalar quadratic-form* oracle. Its exponential lower bound is not a lower bound for the stronger, unrestricted full-vector oracle above. Section 6 of that paper poses Conjecture 23 about projection onto arbitrary spans of product queries. Section 3 gives a counterexample to the exact statement in arXiv v2; it does not contradict the paper’s conditioned theorems.

Meyer–Avron [3] analyze Kronecker–Hutchinson estimators, give a root-mean-square-error lower bound using Wishart factors, and describe exact recovery of tensor-product matrices in $qn + 1$ calls. An estimator-specific variance lower bound is not a minimax lower bound, and an RMSE lower bound alone does not imply the success-probability criterion (1). Our lower bound below supplies a separate anti-concentration argument for that criterion. The parallel access identity improves the stated $qn + 1$ recovery construction to n calls under the PSD-factor promise. No claim of an exhaustive novelty or priority check is made.

1.3 General bounds and the gap

Put

$$\alpha_n = \frac{3n}{n+2}, \quad \beta_n = \frac{2n}{n+1}, \quad (2)$$

$$L(n, q, \varepsilon) = \max_{1 \leq b \leq q} \min \left\{ n^b, \frac{\sqrt{[q/b]}}{\varepsilon} \right\}. \quad (3)$$

Define $U(n, q, \varepsilon)$ to be the minimum of the following four integers:

$$U_0 = n^q, \quad (4)$$

$$U_{\mathbb{R}} = \left\lceil \frac{3(\alpha_n^q - 1)}{\varepsilon^2} \right\rceil, \quad (5)$$

$$U_{\mathbb{C} \rightarrow \mathbb{R}} = (q+1) \left\lceil \frac{3(\beta_n^q - 1)}{\varepsilon^2} \right\rceil, \quad (6)$$

$$U_{\text{ordinary}} = n^{q-1} (5 \lceil 4/\varepsilon \rceil + 4). \quad (7)$$

Theorem 1.1 (General bounds). *For the full RA-11 model,*

$$\frac{1}{80000} L(n, q, \varepsilon) \leq Q(n, q, \varepsilon) \leq U(n, q, \varepsilon). \quad (8)$$

In particular, $Q(n, q, \varepsilon) = \Theta(n^q)$ in the parameter regime $0 < \varepsilon \leq n^{-q}$, with universal constants.

The upper bound follows from Sections 2, 4, and 7; the lower bound follows from Sections 8 and 9. Taking $b = q$ in (3) proves the last assertion together with U_0 .

Unresolved in this note. Away from such special regimes, (8) can have a very large gap. Neither a matching upper bound for L nor a matching lower bound for U is established. Therefore Theorem 1.1 does not determine Q up to universal constants simultaneously in all three parameters.

1.4 A subclass that is characterized

For $n \geq 2$ and $k \geq 1$, let $Q_{\text{prod}}(n, k, \varepsilon)$ denote the same minimax quantity with the promise

$$M = A_1 \otimes \cdots \otimes A_k, \quad A_i \in \mathbb{R}^{n \times n}, \quad A_i \succeq 0. \quad (9)$$

The local factors are unknown and are not supplied to the algorithm.

Theorem 1.2 (Tensor-product subclass). *For all these parameters,*

$$\frac{1}{80000} \min \left\{ n, \frac{\sqrt{k}}{\varepsilon} \right\} \leq Q_{\text{prod}}(n, k, \varepsilon) \quad (10)$$

$$\leq \min \left\{ n, 5 \lceil 8\sqrt{k}/\varepsilon \rceil + 5 \right\} \leq 45 \min \left\{ n, \frac{\sqrt{k}}{\varepsilon} \right\}. \quad (11)$$

The n -query upper bound recovers the entire matrix in factored form, and its trace exactly, with probability one for every fixed promised input.

Only the subclass in (9) has the matching rate above. Low tensor-product complexity of the *queries* does not imply this promise about the *matrix*.

2 Complex product queries from $q + 1$ real calls

Theorem 2.1 (Interpolation simulation). *Let M be any real linear map on $(\mathbb{R}^n)^{\otimes q}$, not necessarily symmetric or PSD. For any $z_i = a_i + ib_i \in \mathbb{C}^n$, the complexified response $M(z_1 \otimes \cdots \otimes z_q)$ is computable using $q + 1$ real product matrix-vector queries and finite exact arithmetic.*

Proof. Consider the vector-valued polynomial

$$p(t) = \bigotimes_{i=1}^q (a_i + tb_i), \quad \deg p \leq q.$$

Choose distinct real numbers $t_0, \dots, t_q$ and set

$$\ell_j(t) = \prod_{\substack{0 \leq h \leq q \\ h \neq j}} \frac{t - t_h}{t_j - t_h}.$$

Coordinatewise Lagrange interpolation, followed by real linearity, gives

$$M \left(\bigotimes_{i=1}^q z_i \right) = Mp(i) = \sum_{j=0}^q \ell_j(i) M \left(\bigotimes_{i=1}^q (a_i + t_j b_i) \right). \quad (12)$$

Every actual oracle input has real factors. Complex numbers on the right are stored as pairs of reals. Integer nodes make all interpolation coefficients Gaussian-rational, so even the coefficients require no limiting operation or root computation. $\square$

The simulation is adaptive: after reconstructing a complex response exactly, one may form the next complex query of an arbitrary adaptive algorithm and repeat. Thus a t -query complex-product algorithm on a real matrix gives a $(q+1)t$ -query real-product algorithm with the same output distribution.

Remark 2.2 (Conditioning and arithmetic scope). The interpolation coefficients may be large, and the real query matrix can be very ill-conditioned. The proof is an exact identity, not a stable numerical algorithm. It neither uses infinitesimal inputs nor extracts digits of unknown real numbers. An expansion with 2^q real product terms is always possible but is not necessary in this unrestricted exact model.

3 An explicit counterexample to the projection conjecture

Conjecture 23 of arXiv:2502.08029v2 [2] asserts an exponentially small projection bound for a random real unit product vector onto a fixed span of polynomially many real product vectors. The particular asserted scale is c_1^{-q}/n^q , with probability at least $1 - c_2^{-q}$, for constants $c_1, c_2 > 1$. The following deterministic inequality contradicts that scale.

Theorem 3.1 (Failure at $n = 2$). *For every $q \geq 2$ there is a fixed $(q+1)$ -dimensional subspace $S \subseteq (\mathbb{R}^2)^{\otimes q}$ spanned by $q+1$ real product vectors such that*

$$\|P_S u\|_2^2 \geq 2^{1-q} \quad (13)$$

for every real unit product vector $u = u_1 \otimes \dots \otimes u_q$.

Proof. Choose distinct real nodes $t_0, \dots, t_q$ and let

$$S = \text{span}_{\mathbb{R}} \{ (e_1 + t_j e_2)^{\otimes q} : 0 \leq j \leq q \}.$$

The coefficient vectors D_h in

$$(e_1 + t e_2)^{\otimes q} = \sum_{h=0}^q t^h D_h$$

have disjoint nonempty supports: D_h is the sum of coordinate tensors with exactly h occurrences of e_2 . They are linearly independent. The Vandermonde matrix at the distinct nodes is nonsingular, hence $\dim S = q+1$.

Set

$$z = (e_1 + ie_2)^{\otimes q}, \quad r = \operatorname{Re} z, \quad s = \operatorname{Im} z.$$

Interpolation places both r and s in S . Their supports have respectively even and odd parity, so

$$r^T s = 0, \quad \|r\|_2^2 = \|s\|_2^2 = 2^{q-1}.$$

Write $u_i = (a_i, b_i)^T$, with $a_i^2 + b_i^2 = 1$. Then

$$(r^T u)^2 + (s^T u)^2 = \left| \prod_{i=1}^q (a_i + ib_i) \right|^2 = \prod_{i=1}^q (a_i^2 + b_i^2) = 1.$$

Projection onto the two-dimensional subspace spanned by r, s therefore has squared norm exactly 2^{1-q} . Projection onto its superspace S can only increase that norm. $\square$

For any $c_1 > 1$,

$$2^{1-q} > c_1^{-q} 2^{-q}.$$

Thus the conjectured upper-bound event has probability zero for this fixed choice of S , rather than probability at least $1 - c_2^{-q} > 0$. The number of queries is $q + 1$, which is polynomial; normalizing each spanning product vector leaves S unchanged. This contradicts the universal polynomial-size-span formulation in the cited v2. It does not exclude restricted statements with an explicit condition-number bound or a substantially smaller allowed span.

Proposition 3.2 (An explicit conditioning calculation). *The same subspace S has a normalized real product spanning family V with*

$$\kappa_2(V)^2 = \binom{q}{\lfloor q/2 \rfloor} \geq \frac{2^q}{q+1}.$$

Thus even unit-norm query columns can give the exponentially ill-conditioned span used by the counterexample.

Proof. Let $L = q + 1$, $\theta_j = \pi j/L$, and $v_j = (\cos \theta_j e_1 + \sin \theta_j e_2)^{\otimes q}$ for $0 \leq j < L$. The Gram matrix has entries

$$(V^T V)_{jh} = \cos^q(\theta_j - \theta_h) = 2^{-q} \sum_{a=0}^q \binom{q}{a} e^{i(q-2a)(\theta_j - \theta_h)}.$$

The matrix $F_{ja} = L^{-1/2} e^{i(q-2a)\theta_j}$ is unitary, by the finite geometric-series identity. Hence the Gram eigenvalues are exactly $L 2^{-q} \binom{q}{a}$, $0 \leq a \leq q$. They are all positive, so the columns span the full symmetric tensor subspace S , and their largest-to-smallest ratio is the stated binomial coefficient. Its lower bound follows because the largest of the $q + 1$ binomial coefficients is at least their average $2^q/(q+1)$. $\square$

4 General-purpose trace-estimation upper bounds

4.1 Real sphere product probes

Let x_i be independent uniform vectors on the real sphere of radius $\sqrt{n}$, and let $x = \bigotimes_i x_i$. Then $\mathbb{E}[xx^T] = I_N$. For a local coefficient vector a ,

$$\mathbb{E}(a^T x_i)^4 = \frac{3n}{n+2} \|a\|_2^4 = \alpha_n \|a\|_2^4.$$

We record an induction that is useful because it requires no assumptions on the eigenvectors of M .

Lemma 4.1. *For every $a \in (\mathbb{R}^n)^{\otimes q}$,*

$$\mathbb{E}|a^T x|^4 \leq \alpha_n^q \|a\|_2^4.$$

Proof. The local identity proves the case $q = 1$. For the induction step, write the contraction with the last $q-1$ factors as $f = (f_1, \dots, f_n)^T$. Conditioning on those factors gives $\mathbb{E}_{x_1}(f^T x_1)^4 = \alpha_n (\sum_j f_j^2)^2$. If a_j is the corresponding coefficient tensor for f_j , Cauchy–Schwarz and the induction hypothesis give

$$\mathbb{E}\left(\sum_j f_j^2\right)^2 \leq \sum_{j,h} (\mathbb{E}f_j^4 \mathbb{E}f_h^4)^{1/2} \leq \alpha_n^{q-1} \left(\sum_j \|a_j\|_2^2\right)^2.$$

Multiplication by the first factor α_n proves the assertion. $\square$

Let $M = \sum_j \lambda_j a_j a_j^T$ be a spectral decomposition with $\lambda_j \geq 0$ and unit a_j , and put $\tau = \text{tr } M$. Cauchy–Schwarz in L^2 , or the triangle inequality in L^2 , gives

$$\mathbb{E}(x^T M x)^2 \leq \left(\sum_j \lambda_j \sqrt{\mathbb{E}(a_j^T x)^4}\right)^2 \leq \alpha_n^q \tau^2.$$

Consequently the average of m independent scalar observations has variance at most $(\alpha_n^q - 1)\tau^2/m$. Chebyshev’s inequality with $m = U_{\mathbb{R}}$ proves (5). Each scalar is computed from one full-vector oracle response. For $M = 0$ the output is exactly zero.

4.2 Complex sphere product probes, implemented with real inputs

Let z_i be independent uniform complex sphere vectors of radius $\sqrt{n}$, let $z = \bigotimes_i z_i$, and put $Z = z^* M z$. The usual partial trace $\text{tr}_S M$ traces out subsystems indexed by S ; for $S = [q]$ its value is the scalar τ , viewed as a 1×1 matrix.

Lemma 4.2 (Exact second moment). *For every Hermitian PSD M ,*

$$\mathbb{E}Z = \tau, \tag{14}$$

$$\mathbb{E}Z^2 = \left(\frac{n}{n+1}\right)^q \sum_{S \subseteq [q]} \|\text{tr}_S M\|_F^2 \leq \beta_n^q \tau^2. \tag{15}$$

The upper bound is attained by a rank-one projector onto a product vector.

Proof. The local second and fourth moments are

$$\mathbb{E}[z_i z_i^*] = I_n, \quad \mathbb{E}[(z_i z_i^*) \otimes (z_i z_i^*)] = \frac{n}{n+1} (I + F_i),$$

where F_i swaps the two copies of local subsystem i . Expanding $\prod_i (I + F_i)$ and contracting with $M \otimes M$ gives (15); the swap indexed by S gives $\text{tr}[(\text{tr}_{S^c} M)^2]$, and complementation only permutes the sum. Every partial trace is PSD and has trace τ , so its squared Frobenius norm is at most τ^2 . There are 2^q terms. For a product rank-one matrix every partial trace is rank one or a scalar, so all these inequalities are equalities. $\square$

This is the complex sphere version of the partial-trace moment method in [3]. Averaging $m = \lceil 3(\beta_n^q - 1)/\varepsilon^2 \rceil$ independent copies and applying Chebyshev gives failure probability at most $1/3$. Theorem 2.1 computes each complex response in $q + 1$ real calls, proving (6) for the real oracle.

No square root is needed to implement sphere normalization. Draw $w_i = a_i + ib_i$ with independent real standard Gaussian entries, simulate $M(\bigotimes_i w_i)$, and return

$$\left(\prod_{i=1}^q \frac{n}{\|w_i\|_2^2} \right) (\bigotimes_i w_i)^* M(\bigotimes_i w_i).$$

This has precisely the same distribution as Z , and the zero-denominator exceptions have probability zero.

4.3 Coordinate and ordinary-query simulations

Querying every coordinate product vector reveals every diagonal entry and gives $U_0 = N$ calls. For an arbitrary vector $g \in \mathbb{R}^{n^q}$, reshape it as a tensor and write

$$g = \sum_{j_2, \dots, j_q=1}^n g_{:,j_2, \dots, j_q} \otimes e_{j_2} \otimes \dots \otimes e_{j_q}. \quad (16)$$

Thus any ordinary matrix-vector product Mg costs at most n^{q-1} product queries. Lemma 7.1 below, with $r = \lceil 4/\varepsilon \rceil$, supplies an ordinary algorithm with $5r + 4$ calls and failure probability at most $1/4$. Combining it with (16) proves (7).

5 Exact recovery under a low-rank PSD promise

Proposition 5.1. *If $M \succeq 0$ has known rank bound r , then r independent generic real product queries recover M exactly with probability one.*

Proof. Let $s = \text{rank } M \leq r$. The images under M of the coordinate product basis span range M , so some s such images are independent. Therefore the determinant of an appropriate $s \times s$ minor of $[Mx_1, \dots, Mx_r]$, considered as a polynomial in all the local factor entries of $x_1, \dots, x_r$, is not the zero polynomial. Independent Gaussian entries avoid its zero set almost surely. Consequently $Y = MX$, where $X = [x_1, \dots, x_r]$, has rank s almost surely.

Select s independent columns Y_J , with corresponding input columns X_J . The matrix $K = X_J^T Y_J$ is positive definite. To verify this and the reconstruction, write $M = CC^T$ with $C \in \mathbb{R}^{N \times s}$ full column rank and set $D = C^T X_J$. Then $Y_J = CD$ and D is invertible. Hence

$$K = D^T D \succ 0, \quad Y_J K^{-1} Y_J^T = CD(D^T D)^{-1} D^T C^T = CC^T = M.$$

For $s = 0$ all responses vanish and the reconstruction is the zero matrix. Rank selection and matrix inversion can be performed by finite exact arithmetic. $\square$

In particular, for a nonzero rank-one PSD matrix and a generic query x with response $y = Mx$,

$$\text{tr } M = \frac{\|y\|_2^2}{x^T y}. \quad (17)$$

This one-query fact is also noted in [3]. It illustrates why a scalar quadratic-form lower bound cannot simply be relabeled a full-vector lower bound.

6 Parallel access to all PSD tensor factors

Assume the product promise (9). The following construction is an exact oracle reduction; it does not require identifying local factors by a numerical tensor decomposition.

Choose independent standard Gaussian reference vectors $g_i \in \mathbb{R}^n$ and query

$$y_0 = M(g_1 \otimes \cdots \otimes g_k).$$

Compute the scalar

$$\gamma = (g_1 \otimes \cdots \otimes g_k)^T y_0.$$

For the proof only, define

$$\gamma_i = g_i^T A_i g_i, \quad B_i = A_i / \gamma_i. \quad (18)$$

If $M \neq 0$, every A_i is nonzero and PSD, so all $\gamma_i > 0$ almost surely. Then

$$\gamma = \prod_i \gamma_i > 0, \quad M = \gamma \bigotimes_i B_i, \quad g_i^T B_i g_i = 1.$$

The individual γ_i need not be known. If $\gamma = 0$, return zero; this is correct on $M = 0$ and the exceptional event has probability zero on every fixed nonzero promised input.

For a tensor y , let $\text{contr}_{-i}(y; g)$ mean contraction of every output mode $j \neq i$ with g_j . Compute

$$b_i = \frac{\text{contr}_{-i}(y_0; g)}{\gamma} = B_i g_i. \quad (19)$$

The equality follows because each omitted contraction contributes γ_j . In particular, $b_i^T g_i = 1$.

Theorem 6.1 (One round of all local products). *After the reference query, any prescribed tuple of vectors $v_1, \dots, v_k \in \mathbb{R}^n$ can be mapped to the entire tuple $(B_1 v_1, \dots, B_k v_k)$ with one further original product query. The vectors can depend on all previous responses.*

Proof. Using the known b_i , define

$$\alpha_i = 1 - b_i^T v_i, \quad w_i = v_i + \alpha_i g_i. \quad (20)$$

Because B_i is symmetric and $b_i = B_i g_i$,

$$g_i^T B_i w_i = b_i^T w_i = b_i^T v_i + \alpha_i b_i^T g_i = 1.$$

Query $y = M(\bigotimes_i w_i)$. All other-mode contractions are now normalized, so

$$\frac{\text{contr}_{-i}(y; g)}{\gamma} = B_i w_i.$$

Subtracting $\alpha_i b_i$ recovers $B_i v_i$. No division by an unknown bilinear form is used, even when $b_i^T v_i = 0$ or $B_i v_i = 0$. $\square$

Corollary 6.2 (Exact n -call recovery). *Every promised matrix (9) can be recovered, in the form $\gamma \bigotimes_i B_i$, using at most n original calls almost surely.*

Proof. One local column $B_i g_i = b_i$ is already known. Complete g_i to a basis using $n - 1$ coordinate vectors. Such a completion can be found by omitting any coordinate e_j for which $(g_i)_j \neq 0$. Use Theorem 6.1 for the $n - 1$ remaining basis columns in parallel across all i . If X_i is the resulting known local basis and $Y_i = B_i X_i$ its known image, $B_i = Y_i X_i^{-1}$. The total is one reference call plus $n - 1$ rounds. Finally, $\text{tr } M = \gamma \prod_i \text{tr } B_i$. $\square$

The promise is essential: for a general PSD matrix, contractions of a response do not isolate independent local operators, and the normalization in Theorem 6.1 is not an oracle reduction for that general matrix.

7 A local unbiased Hutch++ lemma and the product upper bound

The following version of the standard Hutch++ construction [4] is included to make the constants, unbiasedness, and scale-equivariance used in the parallel product estimator explicit.

Lemma 7.1 (Ordinary PSD trace estimation). *Let $A \in \mathbb{R}^{d \times d}$ be PSD and $r \geq 1$ an integer. There is a nonnegative, unbiased estimator $H_r(A)$ using at most $5r + 4$ ordinary matrix-vector calls, such that*

$$\text{Var } H_r(A) \leq \frac{4}{r(r+1)} (\text{tr } A)^2 \leq \frac{4}{r^2} (\text{tr } A)^2. \quad (21)$$

It can be chosen scale-equivariant for every fixed realization of its randomness: $H_r(cA) = cH_r(A)$ for every $c > 0$.

Proof. If $r \geq d$, query a basis and return the exact trace. Otherwise let $m = 2r + 2$, draw a standard Gaussian $d \times m$ matrix Ω , query $Y = A\Omega$, and let P be the orthogonal projection onto range Y . These are m calls. Using at most m further calls, compute $\text{tr}(PA)$ exactly. For example, choose independent columns F of Y and use

$$P = F(F^T F)^{-1} F^T, \quad \text{tr}(PA) = \text{tr}((F^T F)^{-1} F^T A F).$$

This formula avoids a square-root operation in the implementation.

Let $R = (I - P)A(I - P) \succeq 0$ and draw r fresh independent standard Gaussian vectors h_ℓ . With one call to $A(I - P)h_\ell$ for each, form

$$H_r(A) = \text{tr}(PA) + \frac{1}{r} \sum_{\ell=1}^r h_\ell^T R h_\ell.$$

Conditioned on P , this is nonnegative, has mean $\text{tr } A$, and has variance $2 \|R\|_F^2 / r$. Its total call count is $2m + r = 5r + 4$.

It remains to bound the expected residual norm. Diagonalize A , partition its singular values after the leading r , and split the Gaussian matrix as $\Omega = (\Omega_1^T, \Omega_2^T)^T$, where Ω_1 has r rows. It has full row rank almost surely. The matrix with columns $Y\Omega_1^\dagger [I_r \ 0]$ lies in range Y and agrees with the leading block of A . Its lower-block error shows

$$\|(I - P)A\|_F^2 \leq \sum_{j>r} \lambda_j^2 + \|\Lambda_2 \Omega_2 \Omega_1^\dagger\|_F^2.$$

Conditioning on Ω_1 and averaging Ω_2 makes the second term equal in expectation to $(\sum_{j>r} \lambda_j^2) \|\Omega_1^\dagger\|_F^2$. Moreover,

$$\mathbb{E} \|\Omega_1^\dagger\|_F^2 = \frac{r}{m-r-1} < 1. \quad (22)$$

For completeness, a diagonal entry of $(\Omega_1 \Omega_1^T)^{-1}$ is the reciprocal squared norm of one row projected orthogonally to the other $r-1$ rows. Conditional on those rows the squared norm has distribution χ_{m-r+1}^2 . Integrating its gamma density gives reciprocal mean $1/(m-r-1)$; summing the diagonal proves (22).

Since $\lambda_{r+1} \leq \text{tr } A / (r+1)$,

$$\sum_{j>r} \lambda_j^2 \leq \lambda_{r+1} \sum_{j>r} \lambda_j \leq \frac{(\text{tr } A)^2}{r+1}.$$

Also $\|R\|_F \leq \|(I-P)A\|_F$. Therefore

$$\mathbb{E} \|R\|_F^2 \leq \frac{2(\text{tr } A)^2}{r+1}.$$

Taking the expectation of the conditional variance gives (21); the conditional mean is constant, so there is no additional variance term. Scaling A by $c > 0$ preserves P for fixed Ω and scales both estimator terms by c , proving scale-equivariance. $\square$

For ordinary relative error ε , choosing $r = \lceil 4/\varepsilon \rceil$ gives failure probability at most $1/4$ by Chebyshev, as used earlier.

Proposition 7.2 (Parallel product estimator). *Under (9), at most $5 \lceil 8\sqrt{k}/\varepsilon \rceil + 5$ original queries suffice for relative error ε with probability at least $7/8$.*

Proof. Perform the reference normalization of Section 6 and use Theorem 6.1 to run the local algorithms $H_r(B_i)$ simultaneously, with independent randomness for each factor and $r = \lceil 8\sqrt{k}/\varepsilon \rceil$. They require at most $5r + 4$ rounds after the reference call. Output

$$T = \gamma \prod_{i=1}^k H_r(B_i).$$

Write $\tau_i = \text{tr } A_i$ and first suppose $M \neq 0$. Scale-equivariance gives the pathwise identity

$$\frac{H_r(B_i)}{\text{tr } B_i} = \frac{H_r(A_i)}{\text{tr } A_i}.$$

Thus the normalized local outputs depend only on their independent local algorithmic randomness, not on the shared reference normalization. They are independent, unbiased for one, and have variances at most $4/r^2$. Since $\text{tr } M = \gamma \prod_i \text{tr } B_i$,

$$\mathbb{E} T = \text{tr } M, \quad \frac{\text{Var } T}{(\text{tr } M)^2} \leq \left(1 + \frac{4}{r^2}\right)^k - 1 \leq e^{4k/r^2} - 1 \leq \frac{\varepsilon^2}{8}.$$

The last inequality uses $4k/r^2 \leq \varepsilon^2/16 < 1/64$ and $e^x - 1 \leq 2x$ for $0 \leq x \leq 1$. Chebyshev gives failure at most $1/8$. The zero input is handled by the reference query. $\square$

Taking the smaller of this algorithm and Corollary 6.2 proves the upper bounds in Theorem 1.2.

8 An adaptive, bounded-probability lower bound

8.1 A stronger parallel oracle

For a lower bound on the product subclass, grant the algorithm a stronger oracle: one round returns the individual vectors $A_i v_i$ for all i , rather than only their tensor product. Permit arbitrary local vectors and adaptation across factors, even within a round. Equivalently, allow at most t ordinary queries per factor, in any adaptive interleaving. A product-oracle algorithm with t calls is simulated in this stronger model by taking the tensor product of the individual responses. A lower bound in the stronger model therefore transfers to the original one.

The hard prior uses independent square Wishart matrices in a known local subspace. No condition-number assumption is imposed on the algorithm's queries.

8.2 Adaptive Wishart posterior

Lemma 8.1 (Residual trace distribution). *Let $W_i = G_i^T G_i$, where the $G_i \in \mathbb{R}^{d \times d}$ have independent standard Gaussian entries, independently over $1 \leq i \leq k$. After any adaptively interleaved transcript with at most $t < d$ ordinary matrix-vector queries to each factor, one can augment the transcript with additional queries so that each factor has t independent queried directions. Conditional on this augmented transcript and all algorithmic randomness,*

$$\text{tr } W_i = a_i + X_i, \quad X_i \stackrel{\text{independent}}{\sim} \chi_{(d-t)^2}^2, \quad (23)$$

where a_i is measurable from the transcript and satisfies $0 \leq a_i \leq \text{tr } W_i$ pathwise. The conditional laws of the X_i do not depend on the transcript.

Proof. We give the Gaussian block calculation and then the adaptive induction. For a fixed t -dimensional queried coordinate subspace, write $G = [G_1 \ G_2]$, with G_1 having t columns. The observations give

$$K = G_1^T G_1 \succ 0, \quad C = G_1^T G_2.$$

Conditional on G_1 , split G_2 into its orthogonal projections onto $\text{range } G_1$ and its orthogonal complement. Gaussian orthogonal projections are independent. The first part is $G_1 K^{-1} C$; in an orthonormal basis for the complement, the second part is a $(d-t) \times (d-t)$ standard Gaussian matrix H . Consequently

$$W = \begin{bmatrix} K & C \\ C^T & C^T K^{-1} C + H^T H \end{bmatrix}, \quad (24)$$

with $H^T H$ independent of (K, C) . This independence also holds after integrating out G_1 , because its conditional law is the same for every G_1 . The trace is the sum of

$$a = \text{tr } K + \text{tr}(C^T K^{-1} C)$$

and $\text{tr}(H^T H) \sim \chi_{(d-t)^2}^2$. Both summands are nonnegative.

To justify adaptivity, maintain the following invariant after s independent queries to a factor. In a transcript-measurable orthonormal basis of its queried span and complement, the still unknown Schur complement has law $H_s^T H_s$, for a $(d-s) \times (d-s)$ standard Gaussian H_s , independently of all current observations and of the other factors' unknown Schur complements. The invariant holds initially. Subtract the known queried-span component from the next vector. If the residual is zero,

its response was already determined and reveals no new information. Otherwise its normalized direction is a measurable function of the current transcript. Conditional on that transcript, rotate the complement to make this direction the first basis vector. Orthogonal invariance leaves the Schur-complement distribution unchanged. The newly observed column is precisely one column of that Schur complement plus known terms. Applying (24) with one queried column removes a row and a column, and leaves a fresh square Wishart Schur complement of order $d - s - 1$, independent of the new observations. Conditional independence for other factors is preserved, since the new response depends only on this factor's residual and the already fixed transcript. This proves the invariant for arbitrary interleavings by induction.

Dependent queries can be answered from prior responses. At the end, query fresh orthogonal directions to pad each local span to dimension t ; this only gives the algorithm additional information. The invariant now proves (23). Coordinate-free, the known nonnegative term is

$$a_i = \text{tr} \left(W_i U_i (U_i^T W_i U_i)^{-1} U_i^T W_i \right),$$

where U_i has orthonormal columns spanning the final queried subspace. It is observable from $W_i U_i$, and the remaining Schur-complement trace is nonnegative. This proves $0 \leq a_i \leq \text{tr} W_i$. $\square$

The augmentation is a device for bounding success from above: an estimator based on the original transcript is still a measurable estimator given the augmented transcript. It is not assumed that the original algorithm actually used all the extra responses.

8.3 Three anti-concentration facts

Lemma 8.2 (Logarithms of shifted chi-squares). *Let $X \sim \chi_\nu^2$, with $\nu \geq 2$, and $a \geq 0$. Then $Y = \log(a + X)$ has a log-concave density and*

$$\text{Var } Y \geq \frac{2\nu}{(a + \nu + 2)^2}. \quad (25)$$

Proof. On its support, its log-density, up to a constant, is

$$y + \left(\frac{\nu}{2} - 1 \right) \log(e^y - a) - \frac{e^y - a}{2}.$$

The second derivative equals

$$- \left(\frac{\nu}{2} - 1 \right) \frac{ae^y}{(e^y - a)^2} - \frac{e^y}{2} \leq 0.$$

The support is an interval; extending the density by zero preserves log-concavity. For the variance, integrating the chi-square density by parts gives

$$\text{Cov}(X, \log(a + X)) = 2\mathbb{E} \frac{X}{a + X}.$$

The boundary terms vanish; the case $a = 0$ follows either directly or by a limit. Cauchy-Schwarz applied to $X = \sqrt{X/(a + X)} \sqrt{X(a + X)}$ yields

$$\mathbb{E} \frac{X}{a + X} \geq \frac{(\mathbb{E} X)^2}{\mathbb{E}[X(a + X)]} = \frac{\nu}{a + \nu + 2}.$$

Since $\text{Var } X = 2\nu$, a second Cauchy-Schwarz inequality, this time for the covariance, proves (25). $\square$

Lemma 8.3 (One-dimensional log-concave density bound). *A log-concave probability density f on the real line with variance $\sigma^2 > 0$ satisfies*

$$\|f\|_\infty \leq \frac{8}{\sigma}.$$

Thus the probability of any interval of length δ is at most $8\delta/\sigma$.

Proof. Let $H = \|f\|_\infty$ and let m be a mode; limiting arguments handle an endpoint mode. Let $a, b \geq 0$ be the distances from m to the two endpoints of the superlevel interval where $f \geq H/e$. Its mass implies $a + b \leq e/H$. Log-concavity implies the tail bound $f(m+t) \leq He^{-t/b}$ for $t \geq b$, and the corresponding left-tail bound with a . A side with zero width contributes zero; a truncated support only reduces the integral. Therefore

$$\int (x-m)^2 f(x) dx \leq H(a^3 + b^3) \left(\frac{1}{3} + \frac{5}{e} \right) \leq \frac{e^3/3 + 5e^2}{H^2} < \frac{64}{H^2}.$$

Here $\int_1^\infty u^2 e^{-u} du = 5/e$. The variance is at most the second moment about m , so $H\sigma \leq 8$. $\square$

We also use the standard fact that a convolution of integrable log-concave functions is log-concave, hence a sum of independent random variables with log-concave densities has a log-concave density. This is a consequence of Prékopa's marginal theorem; see the review [5]. Its hypotheses hold for all the densities in Lemma 8.2.

8.4 Success probability under the hard prior

Proposition 8.4 (Parallel oracle lower bound). *For $n \geq 2$, $k \geq 1$, and $0 < \varepsilon < 1/2$, every randomized algorithm succeeding with probability at least $2/3$ for every tuple of $n \times n$ PSD matrices, when estimating $\prod_i \text{tr } A_i$ from the stronger parallel oracle, needs at least*

$$\frac{1}{80000} \min \left\{ n, \frac{\sqrt{k}}{\varepsilon} \right\}$$

queries per factor in the worst case.

Proof. We first work in dimension $d \geq 4$ and suppose the local query budget is $t \leq d/4$. Fix the algorithm's random seed and use the independent Wishart prior of Lemma 8.1; alternatively include that seed in the conditioning throughout. After augmentation, write

$$\text{tr } W_i = a_i + X_i, \quad X_i \sim \chi_\nu^2, \quad \nu = (d-t)^2.$$

The known a_i are nonnegative and $\mathbb{E} \sum_i a_i \leq kd^2$, because $\mathbb{E} \text{tr } W_i = d^2$. Thus the transcript event

$$\mathcal{G} = \left\{ \sum_i a_i \leq 10kd^2 \right\}$$

has probability at least $9/10$ by Markov. On $\mathcal{G}$, at least $k/2$ of the indices satisfy $a_i \leq 20d^2$. Also $9d^2/16 \leq \nu \leq d^2$. For every such index, Lemma 8.2 gives

$$\text{Var}(\log(a_i + X_i) \mid \text{transcript}) \geq \frac{2\nu}{(a_i + \nu + 2)^2} \geq \frac{1}{500d^2}.$$

For the last inequality it suffices to use $a_i + \nu + 2 \leq 22d^2$ and $2\nu \geq 9d^2/8$.

Conditioned on the augmented transcript, the log of the desired trace product is

$$Z = \log \prod_i \text{tr } W_i = \sum_i \log(a_i + X_i).$$

It has a log-concave density, and on $\mathcal{G}$ its conditional variance is at least $k/(1000d^2)$. Lemma 8.3 therefore bounds its conditional density by

$$8\sqrt{1000} \frac{d}{\sqrt{k}}.$$

For a fixed positive estimate $\hat{T}$, relative-error success requires

$$Z \in \left[\log \frac{\hat{T}}{1+\varepsilon}, \log \frac{\hat{T}}{1-\varepsilon} \right].$$

The interval has length $\log((1+\varepsilon)/(1-\varepsilon)) \leq 3\varepsilon$ for $0 < \varepsilon < 1/2$. A nonpositive estimate has zero success probability under this strictly positive prior. The same bound holds for randomized output after conditioning on all its randomness. Consequently every estimator satisfies

$$\mathbb{P}\{\text{relative-error success}\} \leq \frac{1}{10} + 24\sqrt{1000} \frac{\varepsilon d}{\sqrt{k}}. \quad (26)$$

This is a probability bound, not merely a lower bound on mean-square error.

To obtain a uniform bound in the original local dimension n , put $X = \sqrt{k}/\varepsilon$ and choose

$$d = \min\{n, \lfloor X/10000 \rfloor\}.$$

When $d \geq 4$, embed each $d \times d$ Wishart matrix in the first d coordinates of an $n \times n$ matrix and fill the rest with zeros. The algorithm can be told this support; queries outside it reveal nothing else. For $t \leq d/4$, the right-hand side of (26) is at most

$$\frac{1}{10} + \frac{24\sqrt{1000}}{10000} < 0.176 < \frac{2}{3}.$$

An algorithm that succeeds on every fixed input would have average success at least $2/3$ under this prior, a contradiction. Thus $t > d/4$ is necessary. Since $d \geq \min(n, X)/20000$ when $d \geq 4$, this gives the desired constant.

If $d < 4$, then either $n < 4$ or $X < 40000$, so $\min(n, X) < 40000$. Zero calls cannot succeed for every trace: the required output events for a zero input and for a fixed positive-trace input are disjoint and would each need probability at least $2/3$. Hence at least one call is necessary, which is more than $\min(n, X)/80000$ in this remaining case. The prior and every estimate above are independent of the random seed chosen by the algorithm, so averaging over that seed preserves the argument. $\square$

The stronger parallel oracle simulates the original product oracle, so Proposition 8.4 supplies the lower bound of Theorem 1.2. Together with the upper bound already proved, this establishes the claimed $\Theta(\min(n, \sqrt{k}/\varepsilon))$ subclass rate.

Corollary 8.5 (Exact trace recovery on the subclass is optimally n calls). *If the desired error is exactly zero, a randomized algorithm that recovers the trace of every PSD product input with probability at least $2/3$ must use at least n calls in the worst case. Thus Corollary 6.2 is optimal for exact trace recovery.*

Proof. Use the stronger parallel oracle and the independent square Wishart prior in local dimension n . With $t < n$ queries per factor, Lemma 8.1 leaves independent residuals with distributions $\chi_{(n-t)^2}^2$. These have continuous densities even when $(n-t)^2 = 1$. The conditional product $\prod_i (a_i + X_i)$ is positive and has no atoms. Any estimate determined by the transcript and the algorithm's random seed therefore has conditional probability zero of being exactly equal to the trace product. The average success is zero, contradicting a uniform success guarantee. The argument transfers to the weaker original product oracle. $\square$

9 Blocking and a general lower bound

Proof of the lower bound in Theorem 1.1. Fix $1 \leq b \leq q$ and let $k = \lfloor q/b \rfloor$. Partition the first bk modes into k blocks of size b , leaving $q - bk$ modes. Restrict the general PSD input to

$$M = W_1 \otimes \cdots \otimes W_k \otimes I_{n^{q-bk}}, \quad W_j \in \mathbb{R}^{n^b \times n^b}, \quad W_j \succeq 0.$$

For the lower bound, give the algorithm the individual W_j matrix-vector responses and allow arbitrary, not necessarily product, vectors of length n^b in each block. One original product query is simulated by one round of this stronger parallel oracle, using the within-block product of its original factors. The remaining identity response is known. Relative trace estimation of M is equivalent to estimating $\prod_j \text{tr } W_j$, because their traces differ by the known factor n^{q-bk} .

Proposition 8.4, with local dimension n^b , gives

$$Q(n, q, \varepsilon) \geq \frac{1}{80000} \min \left\{ n^b, \frac{\sqrt{\lfloor q/b \rfloor}}{\varepsilon} \right\}.$$

The same constant works for each b ; taking the maximum proves the claim. $\square$

The case $b = q$ gives the ordinary-query lower scale $\min(N, 1/\varepsilon)$. The case $b = 1$ gives $\min(n, \sqrt{q}/\varepsilon)$. Intermediate blocks are useful when the local dimension is small: they prevent the lower bound from being capped at n merely because the original individual modes are small. None of these steps assumes that an arbitrary PSD input factors across blocks; restricting to such inputs is legitimate only on the *lower-bound* side.

10 Scope checks, reproducibility, and remaining work

10.1 A deterministic obstruction that is not a randomized lower bound

There is a simple deterministic N -call lower bound for arbitrary PSD matrices, but it cannot be substituted for the randomized minimax lower bound. Run a deterministic algorithm on I_N using fewer than N calls, and let S be the span of its queried vectors. For every $c > 0$, the PSD matrix $I_N + cP_{S^\perp}$ gives exactly the same transcript: by induction all adaptive queries stay the same and lie in S . Its trace is $N + c(N - \dim S)$, which can be arbitrarily large. No common answer has fixed relative accuracy for all these traces.

For a randomized algorithm, however, S depends on its seed. Choosing a different adversarial matrix after seeing each seed does not exhibit a single fixed input with large failure probability. The Wishart argument in Section 8 avoids that quantifier error by fixing an input prior independent of the algorithm's randomness.

10.2 What has actually been checked

The package contains mathematical proofs, exact finite algebraic checks using SymPy, and floating-point diagnostics using NumPy. These are different kinds of evidence. None of the finite tests substitutes for the general arguments.

Component	Check performed
Interpolation identity	Exact symbolic vector equality for $n = 2$, $q = 2, \dots, 6$, and $n = 3$, $q = 2, 3$.
Projection counterexample	Exact ranks, parity-vector norms, and rational unit-product projection inequalities for $q = 2, \dots, 8$.
Parallel access	Exact rational contraction and factor-recovery checks, including singular PSD factors; total call count n .
Low-rank reconstruction	Exact PSD Nyström identities in ambient dimension eight, with ranks one through four.
Complex second moment	60,000 Monte Carlo samples for each of three small PSD examples, compared with the exact partial-trace formula.
Adaptive Wishart residual	7,000 adaptive Krylov-query trials, with $d = 10$, $t = 3$, compared with the predicted χ^2_{49} moments.

The Wishart diagnostic produced residual mean 48.98584 and variance 98.48736, versus theoretical values 49 and 98. Its empirical correlation with the known trace part was -0.01174 . Agreement of these moments and a small correlation do *not* prove conditional independence; that claim rests on the adaptive induction in Lemma 8.1. The floating-point complex-query simulations tested through $q = 6$ had relative errors below 2.3×10^{-15} on the recorded instances; no stability claim for larger q follows.

The product Hutch++ numerical example uses $n = 20$, $k = 2$, $r = 2$, and makes 15 calls. Its particular relative error was approximately 0.1363. This is only a branch-execution test; $r = 2$ is *not* the theorem’s conservative accuracy choice for a specified small ε .

10.3 Reproduction

From the package root, with Python, NumPy, and SymPy installed, run

```
python tests/exact_checks.py
OPENBLAS_NUM_THREADS=1 python tests/numerical_checks.py
```

The scripts write machine-readable JSON results in `results/`. The recorded numerical seed is 20260913. The reference implementations are in `src/ra11.py`. Running `./build.sh` rebuilds this PDF with `pdflatex`. The algorithms are written for small dense demonstrations; large tensor dimensions, normalization underflow, and interpolation conditioning are not engineered away.

10.4 Final status

The exact interpolation and parallel-access identities do not rely on numerical experiments. The bounded-probability lower bound explicitly handles arbitrary adaptation and removes the need to

infer probability failure from RMSE alone. The resulting subclass theorem and counterexample are concrete results of this note, subject to independent verification.

The missing theorem is a characterization of $Q(n, q, \varepsilon)$ for *arbitrary* PSD matrices in the full parameter range. The present upper and lower bounds do not close that gap. Consequently this package should not be represented as a complete resolution of RA-11, nor used by itself as grounds to mark that problem solved.

References

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